Idea2Impact – Student Idea Validation & Incubation Management System

A PROJECT REPORT

Submitted by

NIVETHA S

(240701374)

in partial fulfilment of the End Semester Examination for the course

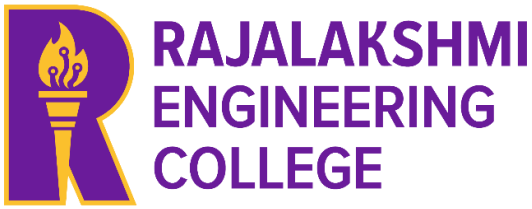
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

[COMPUTER SCIENCE AND ENGINEERING]



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

[NOVEMBER 2026]

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "Idea2Impact – Student Idea Validation & Incubation Management System" is the bonafide work of [NIVETHA S(240701374), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I.NIVETHA S (240701374), hereby declare that the project report titled **"Idea2Impact – Student Idea Validation & Incubation Management**"System.submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Thandalam
Date: 05 November 2026

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ABSTRACT

The **Idea2Impact – Innovation and Incubation Management System** is a web-based application developed using the Zoho Creator low-code platform to simplify and manage the process of idea submission, evaluation, approval, and innovation tracking. The system aims to provide a centralized platform for students and administrators to manage innovative ideas and project proposals efficiently. In the traditional approach, idea submissions and their evaluation may involve manual documentation, spreadsheets, and communication, which can lead to delays, data duplication, and difficulties in tracking the progress of submissions.

The proposed system provides structured forms for submitting and managing innovation ideas, reports for viewing submitted records, and an interactive dashboard for monitoring application activities. It includes status-based record management with categories such as Under Review, Approved, and Rejected. Workflow automation is implemented using Zoho Creator to simplify the evaluation and approval process. Integration with Zoho CRM enables relevant project information to be transferred between Zoho Creator and the CRM platform, improving centralized record management. The project also incorporates an AI integration approach using the OpenAI API to support intelligent analysis of submitted ideas and provide useful insights, subject to successful API configuration.

The application is designed to improve accessibility, data organization, transparency, and efficiency in innovation management. It provides administrators with a centralized view of submissions and their statuses, helping them monitor and manage the overall evaluation process. The application is tested using sample records and functional testing scenarios to verify form submissions, report generation, workflow execution, and integration behaviour. The Idea2Impact system demonstrates how low-code application development, workflow automation, CRM integration, and AI capabilities can be combined to create an efficient innovation management solution.

Keywords: *Zoho Creator, Idea2Impact, Low-Code Development, Innovation Management, Workflow Automation, Zoho CRM Integration, Artificial Intelligence.*

PROJECT AT A GLANCE

> ***Guidance:*** *This one-page profile lets examiners verify the application quickly. Counts must match the live application and Appendix A.*

Attribute	Details
Project Title	Idea2Impact – Innovation and Incubation Management System
Application Domain	Education / Innovation / Research and Development / Incubation Management
Student Name & Register No.	Nivetha S – [240701374]
Zoho Creator Edition	Zoho Creator Trial Edition
Application Name (Link Name)	Idea2Impact
Published / Live URL	https://idea2impact.zoho.in/
Demo Credentials (per role)	Admin
Number of Forms / Subforms	1 confirmed main form – Innovation_Registration_Form
Number of Reports (by type)	4 confirmed views – List/Report: 1. Kanban: 2. Calendar: 3. Spreadsheet: 4. Dashboard charts
Number of Pages / Dashboards	1 confirmed Dashboard
Number of Workflows	3 workflows
Deluge Functions / Custom Actions	1 Standalone scripts for CRM
Integrations	Zoho CRM – Integration workflow configured; successful synchronization pending verification. Email notification configured.
Roles & Profiles Defined	Administrator – application administration
Environments Used	Development / Testing in Zoho Creator
Date of Production Deployment	[DD Month YYYY]

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Idea2Impact – Student Idea Validation and Incubation Management System is developed using Zoho Creator for managing innovation activities at Rajalakshmi Engineering College. It provides a centralized platform for students, faculty, and external innovators to submit their ideas. The system helps administrators manage applications through forms, reports, Kanban boards, and dashboards.

1.2 Problem Statement

The existing innovation management process can involve manual data collection and tracking, making it difficult to manage submissions and monitor their progress. Idea2Impact addresses these challenges through a centralized application management system.

- Difficulty maintaining innovation records.
- Lack of proper application status tracking.
- Time-consuming manual reporting and monitoring**.**

1.3 Objectives

- To create registration forms for submitting innovative ideas.
- To automate application status updates and email notifications.
- To provide reports and dashboards for tracking submissions.
- To connect the application with Zoho CRM for record accessibility.

1.4 Scope of the Project

In Scope:

1. Registration and idea submission for students, faculty, and external innovators.
2. Application tracking through reports, Kanban boards, and dashboards.
3. Email notifications and Zoho CRM integration.

Out of Scope:

1. Automatic AI-based approval or rejection of ideas.
2. Full mobile application and customer portal development.
3. Actual funding disbursement and incubation management.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because its drag-and-drop tools help develop Idea2Impact quickly with minimal coding. It allows citizen developers to create and maintain forms, reports, and workflows easily. Built-in cloud hosting reduces the need for separate server setup and maintenance. It supports web and mobile access, along with security features and user access controls. Its integrated tools reduce development effort and make the application cost-effective.

1.6 Organisation of the Report

Chapter 1 introduces the project. Chapter 2 reviews existing systems and related literature. Chapter 3 presents the requirements. Chapter 4 explains the system design. Chapter 5 describes implementation. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security, and governance. Chapter 8 presents the results and discussion. Chapter 9 concludes the report and suggests future enhancements.

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CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing process, students and innovators submit their ideas through manual forms or separate communication channels. The administrator collects the details and maintains the records. Applications are reviewed manually, and applicants are informed about their status. Tracking submissions and preparing reports can be time-consuming.

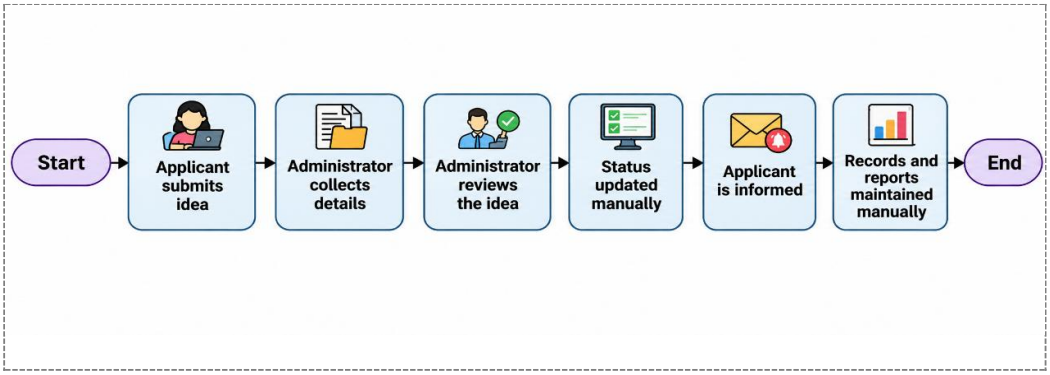


Figure 2.1: [As-Is Process Flow of the Existing System]

2.2 Limitations of the Existing System

- Manual data entry makes record maintenance difficult.
- Tracking application status and review progress is time-consuming.
- Preparing reports and monitoring submissions requires manual effort.

2.3 Low-Code versus Traditional Development

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Takes more Time	Faster Development
Skills required	Programming and coding skills	Basic coding and low-code skills
Hosting & infrastructure	Requires separate hosting setup	Built-in cloud hosting
Mobile application	Requires additional development	Supports mobile access
Security & access control	Must be configured manually	Built-in security and permissions

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Maintenance & change	Requires developer support	Easier to maintain and modify
Customisation limits	Highly flexible	Limited by platform features
Cost	[Generally higher development cost	Generally lower development cost

2.4 Literature Survey

Idea2Impact is a Zoho Creator-based innovation management system that helps students, faculty, and external innovators submit ideas, track reviews, and manage the innovation process.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Bock & Frank (2021)	Low-Code Platform	[Low-code improves development productivity and user participation.]	Supports using Zoho Creator
2	Bucaioni et al. (2022)	Modelling in Low-Code Development	Low-code supports visual modelling and application development.	Supports form and application design.
3	Endres et al. (2022)	Digital Innovation Management	Innovation software helps manage ideas and innovation activities centrally.	Supports centralized idea management.
4	[Zoho Corporation (n.d.)	Zoho Creator Documentation	Creator supports forms, reports, and automated workflows.	Supports registration and status tracking.
5	Jakobsen et al. (2024	Idea Management System	Idea management data can support innovation analysis and AI integration	Supports future AI-based idea evaluation.

2.5 Emerging Trends in Low-Code Development

Low-code development is evolving through AI-assisted app building, citizen development, hyper-automation, and composable applications. These trends influenced Idea2Impact by encouraging faster form creation, easier application maintenance, and automated status updates and email notifications. Zoho Creator enables non-programmers to build applications using visual tools. Future AI integration can help analyse and evaluate submitted innovation ideas.

2.6 Proposed System

[The proposed Idea2Impact system is developed using Zoho Creator to manage innovation submissions in one platform. It reduces manual data entry through centralized registration forms, simplifies tracking through Kanban boards, and provides reports and dashboards for monitoring applications. Automated workflows and email notifications reduce manual effort. Zoho CRM integration improves access to innovation records.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Manages the Idea2Impact application.	Manage registrations, update statuses, view reports and dashboards.
[Applicant	Student, faculty, or external innovator.	Submit ideas and provide required details.
Reviewer	Authorized person who evaluates submitted ideas.	Review applications and update review status.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow applicants to register ideas using the registration form.	Applicant	High
FR-02	The system shall automatically set the initial application status to Submitted.	System	High
FR-03	The system shall send email notifications for application submissions.	Applicant	Medium
FR-04	The system shall allow administrators to update application review statuses.	Administrator	High
FR-05	The system shall display registration statistics through dashboards and reports.	Administrator	Medium
FR-06	The system shall make the Creator application accessible through Zoho CRM.	Administrator	Medium
FR-07	The system shall provide Kanban, spreadsheet, and calendar views of registrations.	Administrator	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports should load within 5 seconds under normal usage.
NFR-02	Security	Users should access only the data permitted by their roles
NFR-03	Usability	The application should be easy to use through a web browser.
NFR-04	Availability	The application should be accessible through Zoho Creator cloud services.
NFR-05	Maintainability	Forms, reports, and workflows should be easy to update
NFR-06	Auditability	Application changes should be traceable wherever audit logging is enabled

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Trial Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman
Other Zoho services	Zoho Creator / Zoho CRM /
Mobile	Zoho Creator mobile app (Android / iOS) [if used]
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

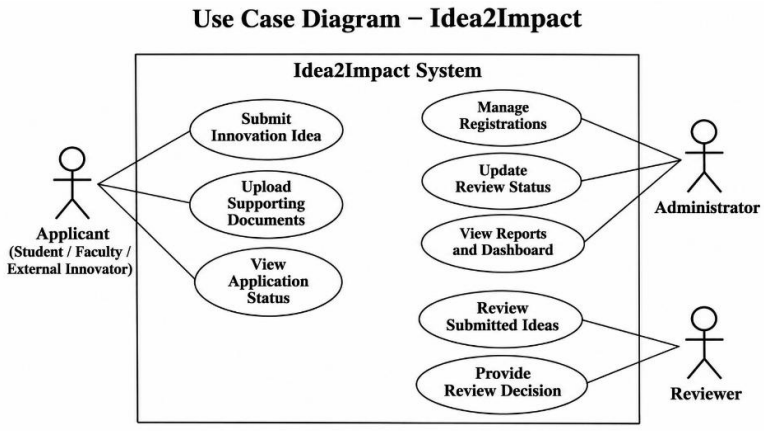


Table 3.5: Use Case Description – [Use Case Name]

Use Case ID	UC-01
Use Case Name	Submit Innovation Idea
Actor(s)	Applicant (Student / Faculty / External Innovator)
Pre-condition	Applicant opens the Innovation Registration Form.
Main Flow	1. Applicant opens the form. 2. Enters required details. 3. Uploads supporting documents. 4. Accepts guidelines. 5. Submits the form.
Alternate Flow	If mandatory fields are missing, the system prompts the applicant to complete them.
Post-condition	Innovation record is saved with the initial status as Submitted.
Related FR	FR-01, FR-02, FR-03

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

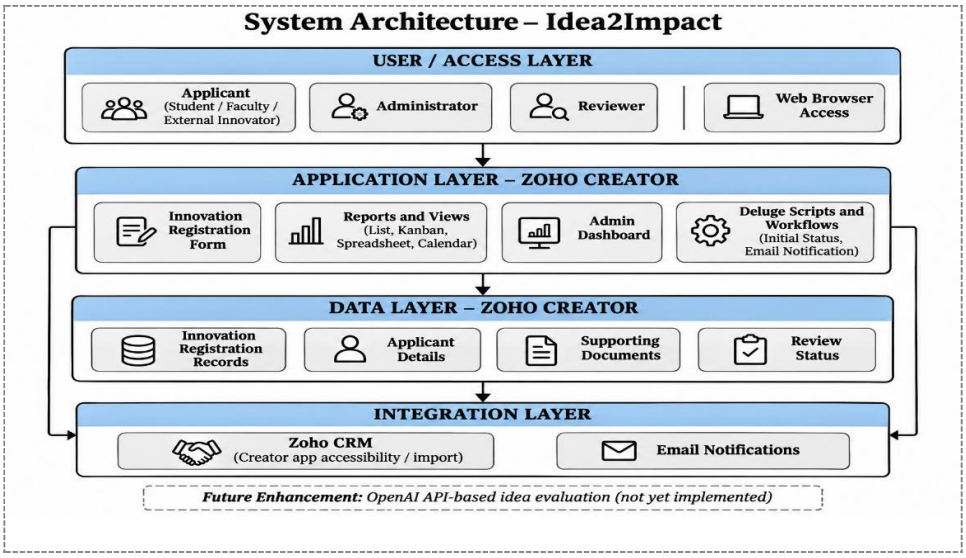


Figure 4.1: System Architecture of the Application

The diagram shows the user layer, Zoho Creator application layer, data layer, and integration layer

4.2 Data Model

Data Model – Idea2Impact

INNOVATION_REGISTRATION_FORM		
Applicant Details	• **Record_ID (PK)** • Applicant_Name • Applicant_Category • Team_Size	• Email_Address • Contact_Number • College_Organization • Institution_Address
Innovation Details	• Innovation_Domain • Technology_Areas • Problem_Statement • Proposed_Solution	• Proposed_Completion_Date • Preferred_Mentor_Meeting_Time • Academic_CGPA
Team & Attachments	• Member_1 • Member_2 • Member_3 • Member_4 • Member_5	• Portfolio_Project_URL • Project_Demo_Video • Idea_Cover_Image • Student_ID_Proof • Special_Requirements_Voice_Note
Review & Submission	• Review_Status • Needs_Incubation_Support • Estimated_Prototype_Budget • Expected_Funding_Support	• Applicant_Digital_Signature • Submission_Date_Time • Agree_to_Innovation_Guidelines

Single-form data model implemented in Zoho Creator

Table 4.1: Entities and Relationships

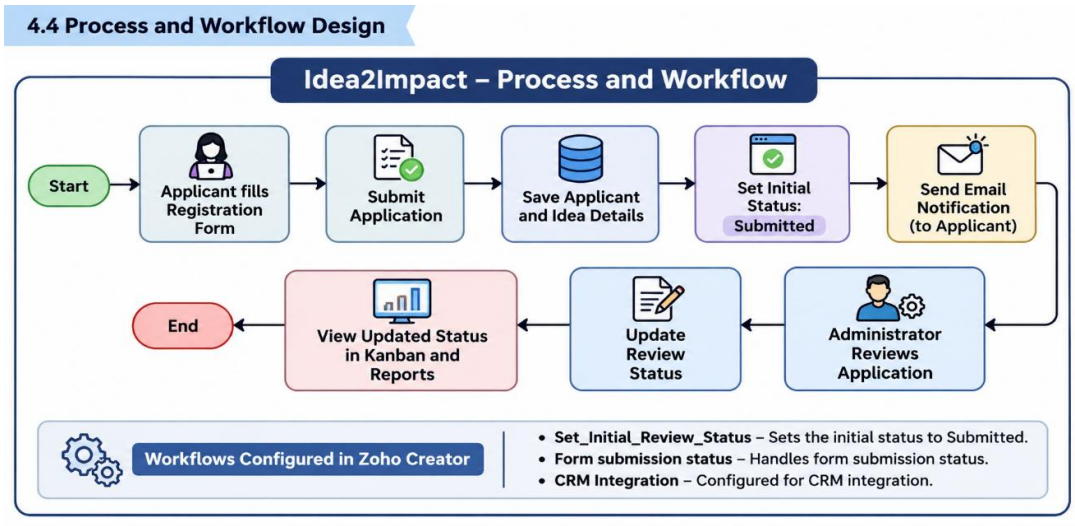
Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Innovation Registration Form	Applicant Name, Email, Contact Number, Innovation Domain, Team Size, Review Status	Applicant Details, Idea Details, Team Details	Single Form	Zoho Creator Form
Team Members	Member 1 to Member 5	Innovation Registration Form	Embedded Fields	Add New Fields
Supporting Documents	Student ID Proof, Idea Cover Image, Demo Video	Innovation Registration Form	One-to-One	File Upload Fields

4.3 Form Design

Table 4.2: Field Specification – [Form Name] Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Applicant Name	Applicant_Name	Name	Yes	First and Last Name	Applicant details
Email Address	Email_Address	Email	Yes	Valid email format	Contact details
Applicant Category	Applicant_Category	Dropdown	[Yes]	Student / Faculty / External Innovator	Applicant type
Team Size	Team_Size	Number	Yes	Maximum 5	Team information
Review Status	Review_Status	Dropdown	Yes	Default: Submitted	Application tracking

4.4 Process and Workflow Design



4.5 Navigation Structure and Wireframe

The diagram below shows the main navigation options and a sample dashboard layout for your application.

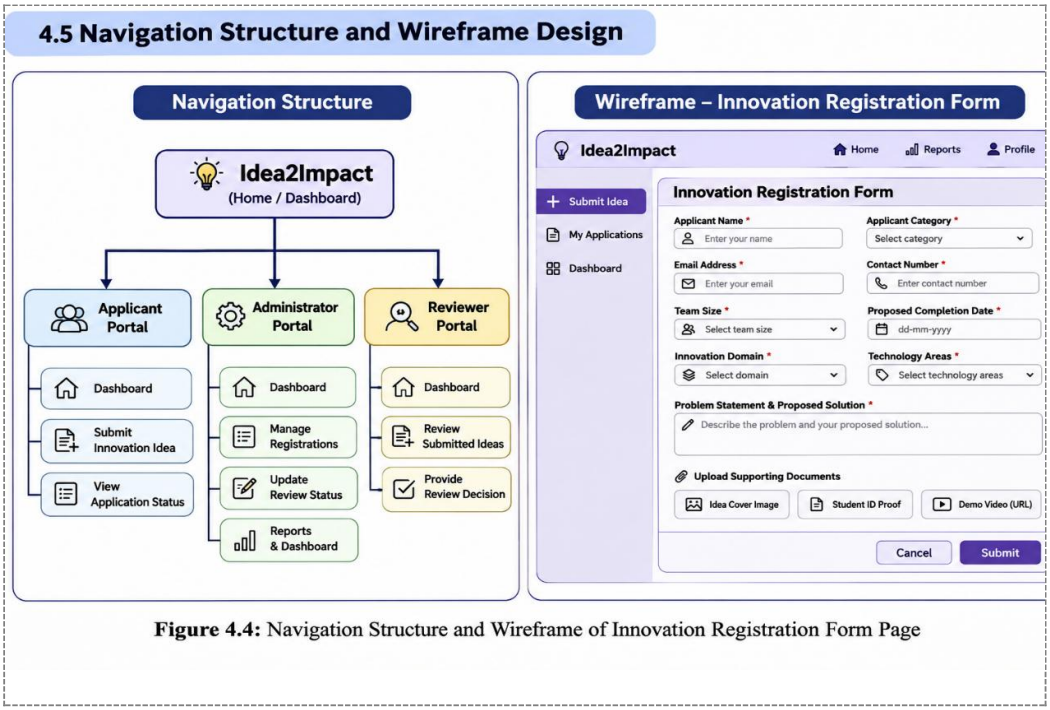


Figure 4.4: Navigation Structure and Wireframe of Innovation Registration Form Page

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Applicant	Reviewer
Innovation Registration Form	A E V D	V	A E V
Registration Report	A E V D	E V	A V
Review Status / Kanban	V	V	V (own)
Admin Dashboard	V	-	—
Zoho CRM Access	V	-	-

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Innovation Registration Form, reports, dashboards, and workflows were configured to manage innovation submissions.

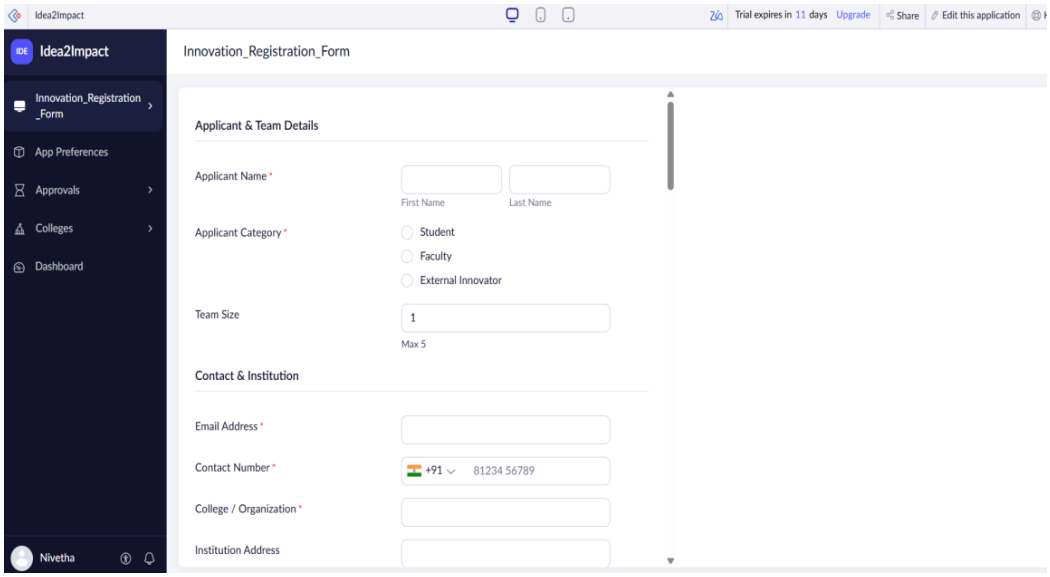


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

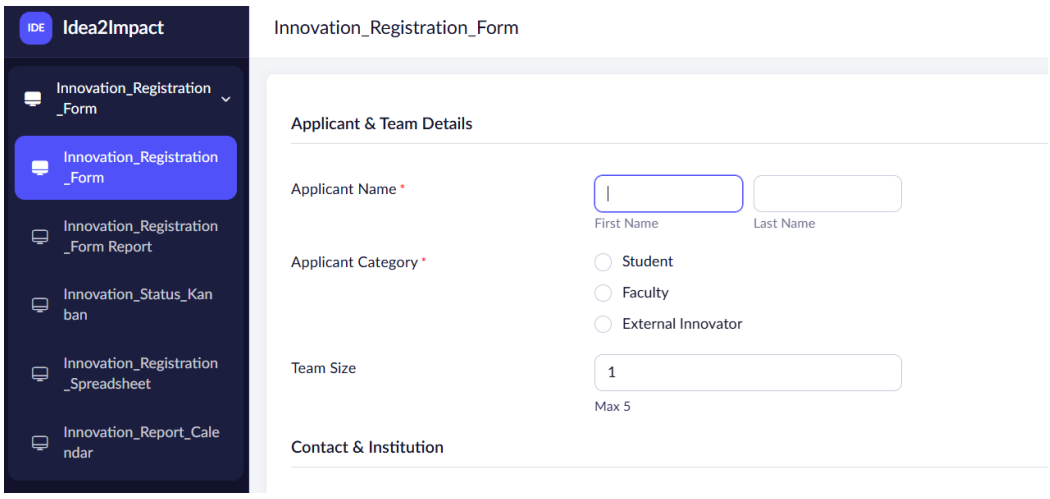


Figure 5.2: Form in the Form Builder



Figure 5.3: Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Innovation Registration Report	[List]	Innovation Registration Form	Displays submitted registration details	-	[Admin, Manager]
Innovation Status Kanban	[Kanban]	Innovation Registration Form	Grouped by Review Status	-	Administrator
Innovation Registration Spreadsheet	Spreadsheet	Innovation Registration Form	Displays records in rows and columns	-	Administrator
Innovation Report Calendar	Calendar	Innovation Registration Form	Month / Week / Day views	-	Administrator

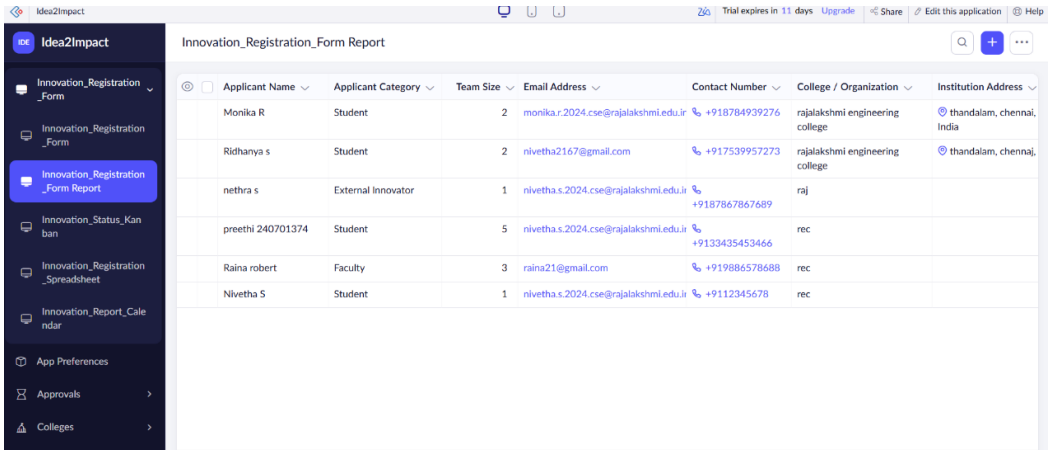


Figure 5.4: [Report Name] Report

5.4 Pages and Dashboards

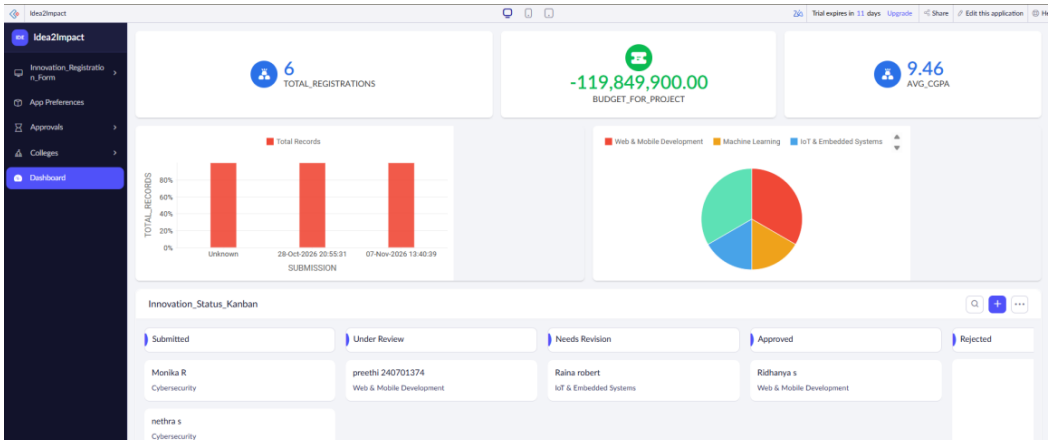


Figure 5.5: Dashboard Page Name

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
[DS-01]	Form Submission	Set initial review status to Submitted	Field assignment
[DS-02]	Form Submission	Handle application submission status	Conditional statements
[DS-03]	CRM Integration Workflow	Support CRM integration	Integration / invokeUrl (if used)

Code 5.1: [Validation script]

```
if(input.Team_Size > 5)
{
   alert "Team size cannot exceed 5 members.";
   cancel submit;
}
```

Code 5.2: [Notification script]

```
sendmail
[
    from : zoho.adminuserid
    to : zoho.adminuserid
    subject : "New Innovation Idea Submitted - " +
input.Applicant_Name
    message : "A new innovation idea has been submitted by " +
input.Applicant_Name +
              ". Please review the application in Idea2Impact."
]
```

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Set_Initial_Review_Status	Form Workflow	On form submission	New record submitted	Set status to Submitted	[DS-01]
Form submission status	Form Workflow	On form submission	-	Update submission status	DS-02
CRM Integration	Integration Workflow	As configured	-	CRM integration action	DS-03

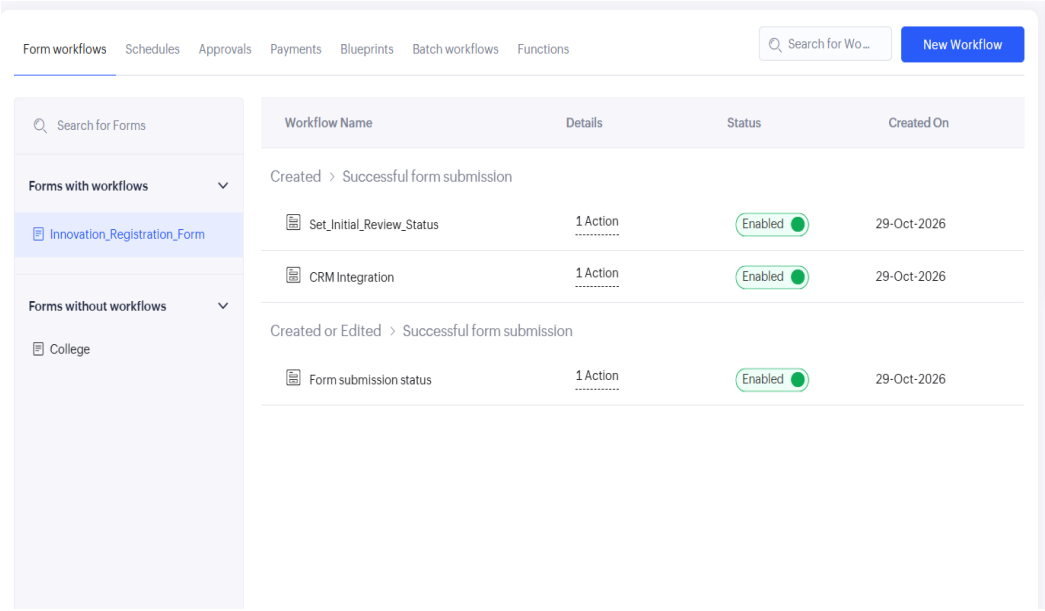


Figure 5.6: [Workflow / Blueprint] Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho CRM	Zoho CRM – Idea2Impact Custom Tab	Import App from Zoho Creator	Zoho Account	Access Creator application through CRM
Email Notification	Zoho Creator Email Service	Workflow Email Action	Platform Managed	Send application submission notifications

Code 5.3: [Integration script – replace with your own]

```
// Integration: Idea2Impact with Zoho CRM
// Method: Import App from Zoho Creator
// Purpose: Make the Creator application accessible
// through the Idea2Impact custom tab in Zoho CRM.

// No custom Deluge invokeurl script used.
```

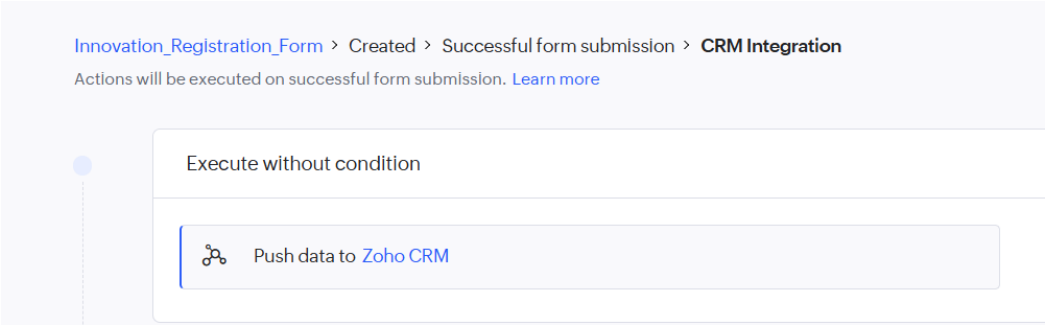


Figure 5.7: Connection Setup and Postman / Response Output

5.8 User Management and Role-Based Access

Users access Idea2Impact through their registered Zoho accounts, with the administrator managing application records and workflows.
Role-based permissions are intended to restrict access to application data and administrative functions.

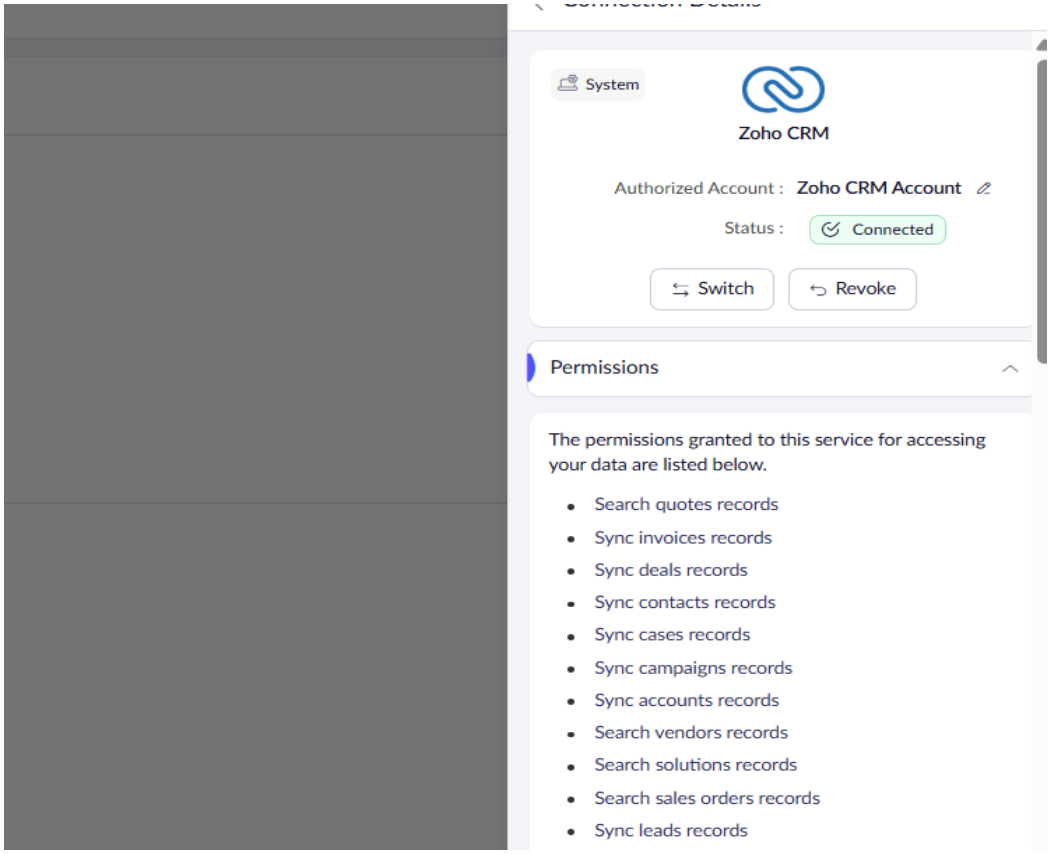


Figure 5.8: [Roles and Permission Profiles Configuration]

5.10 Pro-Code Extension

The Idea2Impact application was tested to check the functionality of registration forms, reports, and workflows. Form testing focused on entering and submitting applicant details. Workflow testing checked initial status updates and email notifications. Reports and dashboards were checked to ensure submitted records were displayed correctly. The application features were reviewed from the administrator's perspective to verify proper operation.

5.10 Application on Mobile

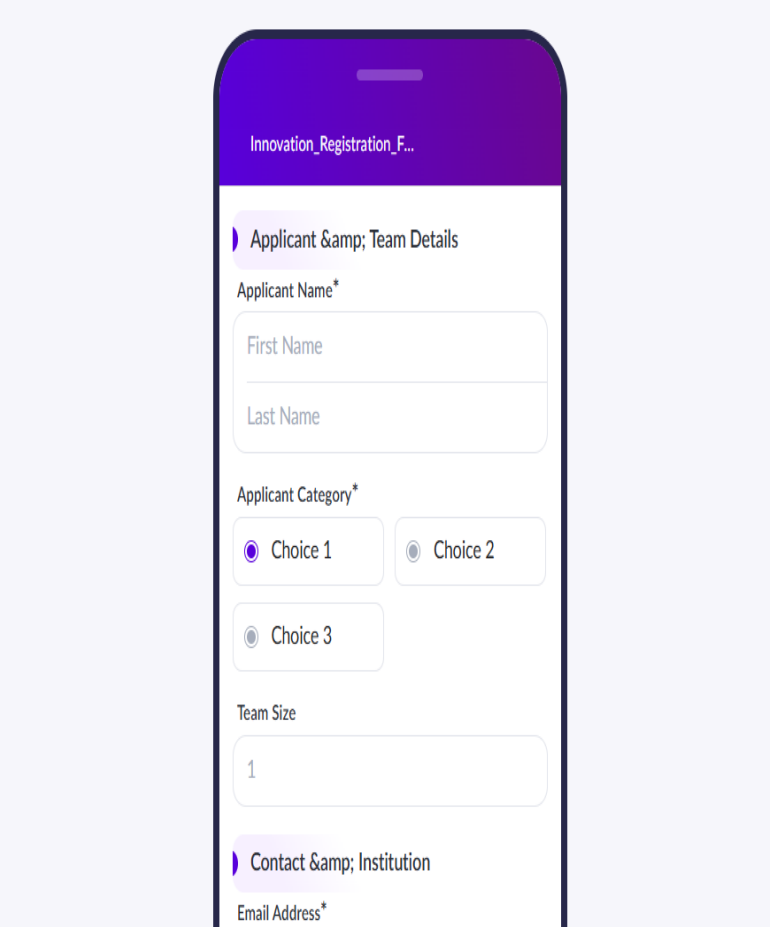


Figure 5.10: [Application on Mobile / Customer Portal]

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

The Idea2Impact application was tested to check the functionality of registration forms, reports, and workflows. Form testing focused on entering and submitting applicant details. Workflow testing checked initial status updates and email notifications. Reports and dashboards were checked to ensure submitted records were displayed correctly. The application features were reviewed from the administrator's perspective to verify proper operation.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	[FR-01]	Submit valid registration	Enter all mandatory details and submit.	Record created successfully.	To be recorded	Pass
TC-02	[FR-01]	Submit form with missing email	Leave email field empty and submit.	Validation error displayed.	To be recorded	Pass
TC-03	[FR-04]	Update review status	Change status to Under Review.	Updated status saved.	To be recorded	Pass
TC-04	[NFR-02]	To be recorded	Open Innovation Status Kanban.	Records grouped by status.	To be recorded	Pass

6.3 Debugging Deluge Scripts

Debugging was performed by checking workflow configurations, execution results, and error messages during form submission. The Deluge info statement helps display field values and trace script execution. Workflow logs can be checked to identify errors in automated actions.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Initial review status not updated	Workflow configuration issue	Corrected workflow configuration	Closed
BUG-02	Invalid team size accepted	Missing validation	Added team size validation	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Applicant	Registration form is easy to use.	No changes required.
User 2	Administrator	Dashboard helps track applications.	No changes required.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Idea2Impact was developed and configured using the Zoho Creator cloud platform. The application is accessed through a web browser using a registered Zoho account. The registration form, reports, dashboards, and workflows were configured and tested in the Creator environment. The application can be accessed by authorized users through the web interface.

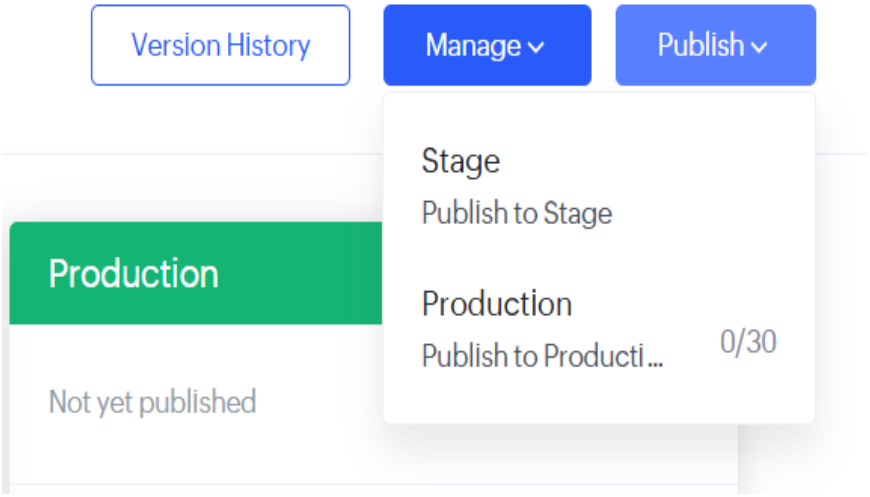


Figure 7.1: [Environments / Publish to Production]

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	✓
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

Centralized record managementIdea2Impact uses Zoho Creator reports and workflows to manage registration records efficiently. Kanban, spreadsheet, and calendar views help organize and access application details. Reports and dashboards provide a centralized view of submissions.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Centralized record management	Registration Form	Manual records	Centralized records	Easier tracking
Status-based organization	Status-based organization	Manual tracking	Status-wise view	Faster monitoring
Dashboard reporting	Admin Dashboard	Manual analysis	Manual analysis	Easier analysis

7.3 Security Features

Idea2Impact uses Zoho Creator's account-based access to provide controlled application access. Applicant details and uploaded documents are maintained within the application. Administrative access is intended for authorized users to manage registrations and review statuses..

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho account authentication	Application login	Unauthorized login
User access permissions	Application access	Unauthorized data access
Mandatory field validation	Registration Form	Incomplete or invalid submissions
Platform-managed data security	Zoho Creator	Data exposure risks

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The Idea2Impact application was tested using a sample innovation registration scenario. An applicant submits an innovative idea through the registration form. The submitted details are stored and displayed in the registration report. The application status is initially set to Submitted, and the administrator can monitor and update the review status using the Kanban board. The dashboard displays registration statistics for easy monitoring.

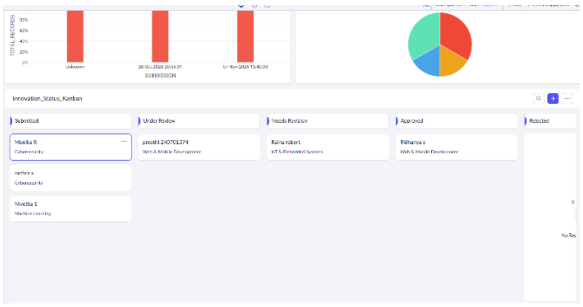


Figure 8.1: Scenario Step 1

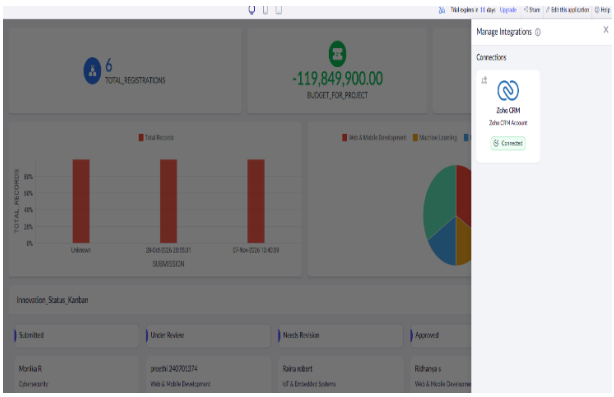


Figure 8.2: Scenario Step 2

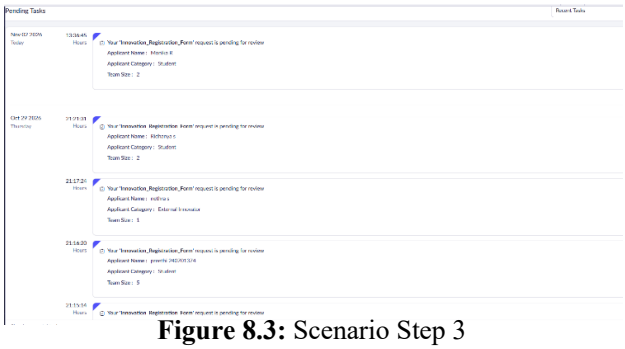


Figure 8.3: Scenario Step 3

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Create registration forms for innovative ideas.	Section 5.2 – Forms	Innovation Registration Form created.
Automate application status updates and notifications.	Section 5.6 – Workflows	Initial status and email workflows configured.
Integrate with Zoho CRM.	Section 5.7 – Integrations	Creator application imported into Zoho CRM.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
[Time to process one request]	Manual processing	Centralized form submission	Reduced manual effort
[Manual steps]	Multiple manual activities	Automated status updates and notifications	Less manual work
[Report generation]	Manual record preparation	Built-in reports and dashboards	Easier reporting
[Data accessibility]	Records maintained separately	Centralized application records	Easier data access

8.4 Limitations of the Low-Code Approach

Zoho Creator provides easy application development, but it has some limitations. Advanced UI customisation is restricted compared to traditional programming. Complex business logic may require additional Deluge scripting. API integrations may also be subject to platform limits.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Idea2Impact application was successfully developed using Zoho Creator to simplify innovation idea submission and management. It provides registration forms, reports, Kanban boards, calendar views, and an administrator dashboard to monitor applications. Workflows and email notifications help reduce manual effort and improve application tracking.

The project demonstrates how low-code development can be used to build an efficient innovation management system. Zoho CRM integration also provides access to the application within the CRM environment**.**

9.2 Future Enhancements

- Integrate the OpenAI API for AI-based analysis and evaluation of submitted ideas.
- Develop a dedicated mobile application and applicant portal for easier access.
- Add advanced analytics to identify promising ideas and monitor innovation progress**.**

9.3 Learning Outcomes and Career Relevance

This project provided practical experience in developing applications using Zoho Creator, designing forms, creating reports, and configuring workflows. I gained knowledge of Deluge scripting, application integration, data management, and dashboard creation. These skills are useful for career paths such as low-code developer, citizen developer, solutions consultant, and automation engineer. The project also improved my problem-solving and application development skills.

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[3] N. Prinz, C. Rentrop, and M. Huber, “Low-Code Development Platforms – A Literature Review,” in *Proceedings of the Americas Conference on Information Systems (AMCIS)*, 2021.

[4] K. Rokis and M. Kirikova, “Exploring Low-Code Development: A Comprehensive Literature Review,” *Complex Systems Informatics and Modeling Quarterly*, no. 36, pp. 68–86, 2023, doi: 10.7250/csimq.2023-36.04.

[5] M. O. Ajimati, N. Carroll, and M. Maher, “Adoption of Low-Code and No-Code Development: A Systematic Literature Review and Future Research Agenda,” *Journal of Systems and Software*, vol. 222, Art. no. 112300, 2025, doi: 10.1016/j.jss.2024.112300.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Idea2Impact	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Innovation Registration Form	5.2	Fig. 5.2
II	Lookup field (relationship between forms)	[Y / N]	College Name Lookup	4.2, 5.2	4.2, 5.2
II	Subform (parent–child data)	N	Not implemented	4.2, 5.2	-
II	Field validations / layout rules	Y*	Innovation Registration Form	5.2	Fig. 5.2
II	List report with filter, grouping, custom action	N*	Innovation Registration Report	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	Y	Innovation Status Kanban, Calendar	5.3	Fig. 5.4
II	Pivot table / chart report	Y	Admin Dashboard charts	5.3	Fig. 5.5
II	Page / dashboard built with UI Builder	Y	Admin Dashboard	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	N*	No confirmed custom script	5.5	-
II	Deluge – functions, fetch, loops, collections	N	Not implemented	5.5	-
II	Email / in-app notifications	Y	Email Notification Workflow	5.5, 5.6	-
II	Form workflow	Y	Set_Initial_Review_Status	5.6	-
II	Scheduled workflow	N	Not implemented	5.6	-
II	Approval process	N	Not implemented	5.6	-
II	Blueprint	N	Not implemented	5.6	-
II	Batch / report workflow	N	Not implemented	5.6	-

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	High	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	Medium	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	Medium	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Low	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	Medium	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Set Initial Review Status

Script ID: DS-01
Location: Innovation Registration Form – On Success
Purpose: Set the initial review status after a new application is submitted.

```
DS-01: Set Initial Review Status

input.Review_Status = "Submitted";

info "Initial review status set to Submitted"
```

C.2 Display Applicant Information

Script ID: DS-02
Location: Innovation Registration Form – On Success
Purpose: Display applicant information in the execution log for debugging.

```
DS-02: Applicant Information Debugging

info "Applicant Name: " + input.Applicant_Name;

info "Email Address: " + input.Email_Address;

info "Application submitted successfully";
```

C.3 Validate Team Size

Script ID: DS-03
Location: Innovation Registration Form – Validation
Purpose: Prevent applicants from entering a team size greater than five.

```
DS-03: Team Size Validation

if(input.Team_Size > 5)

{

   alert "Team size cannot exceed 5 members.";

   cancel submit;

}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

- ✓ Open Google Chrome or another web browser.
- Visit the Zoho Creator website: https://www.zoho.com/creator/.
- Log in using your registered Zoho account.
- Open the Idea2Impact application.
- For mobile access, use the Zoho Creator mobile app and log in with your Zoho account.

D.2 Instructions for [Role 1]

- Open the Innovation Registration Form.
- Enter personal, academic, and innovation details.
- Upload the required documents and idea cover image.
- Accept the innovation guidelines and submit the application.
- Check the application details and status through the available reports.

D.3 instructions for [Role 2]

- Log in and open the Idea2Impact application.
- Open the Innovation Registration Report to view submitted applications.
- Use the Innovation Status Kanban to monitor review stages.
- Update the application status when required.
- Open the Admin Dashboard to view registration statistics and charts.
- Use the spreadsheet and calendar reports to monitor applications.

APPENDIX E – EVALUATION RUBRIC

Student Name	S NIVETHA	Register No.	240701374
Project Title	IDEA2IMPACT	Date	05-11-2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	S NIVETHA	Register No.	240701374
Department / Section	CSE	Date of Viva	05-11-2026
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____